

4.54 Let $Y = \prod_{i=1}^n X_i$. Then $P(Y \leq y) = P(\prod_{i=1}^n X_i \leq y) = P(\sum_{i=1}^n -\log X_i \geq -\log y)$. Now, $-\log X_i \sim \text{exponential}(1) = \text{gamma}(1, 1)$. By Example 4.6.8, $\sum_{i=1}^n -\log X_i \sim \text{gamma}(n, 1)$. Therefore,

$$P(Y \leq y) = \int_{-\log y}^{\infty} \frac{1}{\Gamma(n)} z^{n-1} e^{-z} dz,$$

and

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} \int_{-\log y}^{\infty} \frac{1}{\Gamma(n)} z^{n-1} e^{-z} dz \\ &= -\frac{1}{\Gamma(n)} (-\log y)^{n-1} e^{-(-\log y)} \frac{d}{dy} (-\log y) \\ &= \frac{1}{\Gamma(n)} (-\log y)^{n-1}, \quad 0 < y < 1. \end{aligned}$$

4.55 Let X_1, X_2, X_3 be independent $\text{exponential}(\lambda)$ random variables, and let $Y = \max(X_1, X_2, X_3)$, the lifetime of the system. Then

$$\begin{aligned} P(Y \leq y) &= P(\max(X_1, X_2, X_3) \leq y) \\ &= P(X_1 \leq y \text{ and } X_2 \leq y \text{ and } X_3 \leq y) \\ &= P(X_1 \leq y)P(X_2 \leq y)P(X_3 \leq y). \end{aligned}$$

by the independence of X_1, X_2 and X_3 . Now each probability is $P(X_1 \leq y) = \int_0^y \frac{1}{\lambda} e^{-x/\lambda} dx = 1 - e^{-y/\lambda}$, so

$$P(Y \leq y) = \left(1 - e^{-y/\lambda}\right)^3, \quad 0 < y < \infty,$$

and the pdf is

$$f_Y(y) = \begin{cases} 3(1 - e^{-y/\lambda})^2 e^{-y/\lambda} & y > 0 \\ 0 & y \leq 0. \end{cases}$$

3. (a) we'll get

$$\begin{aligned}
 M_Y(t) &= E(e^{t^T y}) = E E(e^{t^T y} | w) \\
 &= E \left[e^{t^T w + t^T (\tau^2 I) t / 2} \right] \\
 &= e^{t^T \tau^2 I t / 2} E(e^{t^T w}) \\
 &= e^{t^T (\tau^2 I) t / 2} e^{t^T \mu + t^T \Sigma t / 2} \\
 &= e^{t^T \mu + t^T (\Sigma + \tau^2 I) t / 2}
 \end{aligned}$$

$$\Rightarrow Y \sim N(\mu, \Sigma + \tau^2 I)$$

(b) let a be a vector with 0's except for a single 1 in the i th element $a_j = \begin{cases} 1 & j=i \\ 0 & j \neq i \end{cases}$

Using the MGF in (a),

$$\begin{aligned}
 E(e^{y_i}) &= E(e^{a^T y}) = M_Y(a^{(i)}) \\
 &= e^{a^T \mu + a^T (\Sigma + \tau^2 I) a} = e^{\mu + (\Sigma_{ii} + \tau^2)/2}
 \end{aligned}$$

where Σ_{ii} is the i th row + i th column of Σ .

(c)

$$\text{Cov}[e^{y_i}, e^{y_j}] = E[e^{y_i} e^{y_j}] - E[e^{y_i}] E[e^{y_j}]$$

$$\begin{aligned}
 E[e^{y_i} e^{y_j}] &= E[e^{y_i + y_j}] \\
 &= E[e^{b^T y}] \text{ where } b \text{ is a vector } b_k = \begin{cases} 1 & k=i, j \\ 0 & k \neq i, j \end{cases} \\
 &= M_Y(b) = e^{b^T \mu + b^T (\Sigma + \tau^2 I) b / 2} \\
 &= \exp\{2\mu + (\Sigma_{ii} + \Sigma_{jj} + 2\Sigma_{ij} + 2\tau^2)/2\}. \text{ So}
 \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(y_i, y_j) &= \exp\{2\mu + (\Sigma_{ii} + \Sigma_{jj} + 2\tau^2)/2 + \Sigma_{ij}\} - e^{\mu + (\Sigma_{ii} + \tau^2)/2} e^{\mu + (\Sigma_{jj} + \tau^2)/2} \\
 &= \exp\{2\mu + (\Sigma_{ii} + \Sigma_{jj} + 2\tau^2)/2 + \Sigma_{ij}\} - e^{2\mu + (\Sigma_{ii} + \Sigma_{jj} + 2\tau^2)/2} \\
 &= e^{2\mu + (\Sigma_{ii} + \Sigma_{jj} + 2\tau^2)/2} (e^{\Sigma_{ij}} - 1)
 \end{aligned}$$

4.

9) a. From (5) $E(e^{w_i}) = e^{0 + \Sigma_{ii}/2}$

$$\begin{aligned} E(y_i) &= E E(y_i | \lambda_i) \\ &= E(\lambda_i) \\ &= E(e^{w_i}) = e^{\Sigma_{ii}/2} \end{aligned}$$

$$\begin{aligned} \text{Cov}(y_i, y_j) &= E(y_i y_j) - E(y_i) E(y_j) \\ &= E E(y_i y_j | \underline{w}) - e^{\Sigma_{ii}/2} e^{\Sigma_{jj}/2} \\ &= E(E(y_i | \underline{w}) E(y_j | \underline{w})) - e^{\Sigma_{ii}/2} e^{\Sigma_{jj}/2} \\ &= E(e^{w_i} e^{w_j}) - e^{\Sigma_{ii}/2} e^{\Sigma_{jj}/2} \\ &= \exp\left\{\left(\Sigma_{ii} + \Sigma_{jj} + 2\Sigma_{ij}\right)/2\right\} - e^{\Sigma_{ii}/2} e^{\Sigma_{jj}/2} \\ &= e^{\Sigma_{ii}/2} e^{\Sigma_{jj}/2} (e^{\Sigma_{ij}} - 1) \end{aligned}$$

5.1 Let $X = \#$ color blind people in a sample of size n . Then $X \sim \text{binomial}(n, p)$, where $p = .01$. The probability that a sample contains a color blind person is $P(X > 0) = 1 - P(X = 0)$, where $P(X = 0) = \binom{n}{0}(.01)^0(.99)^n = .99^n$. Thus,

$$P(X > 0) = 1 - .99^n > .95 \Leftrightarrow n > \log(.05)/\log(.99) \approx 299.$$

5.2

$Y = \#$ of years until X_i is exceeded

$$Y \in \{1, 2, 3, 4, \dots\}$$

$$\{Y=1\} = \{X_2 > X_1\} \quad P(X_2$$

$$Y=2 \Rightarrow \{X_2 < X_1, X_3 > X_1\}$$

$$Y=3 \Rightarrow \{X_2, X_3 < X_1, X_4 > X_1\} \quad \text{if } F(\cdot)$$

$$P(Y=1) = \int_0^\infty P(X_1=x, X_2>x) dx$$

$$= \int_0^\infty f(x) (1-F(x)) dx \quad \star$$

$$P(Y=2) = \int_0^\infty P(X_1=x) P(X_2<x) P(X_3>x) dx \quad \star$$

$$= \int_0^\infty f(x) F(x) (1-F(x)) dx \quad \star$$

$$P(Y=3) = \int_0^\infty f(x) F(x)^2 (1-F(x)) dx$$

$$P(Y=i) = \int_0^\infty f(x) (1-F(x)) [F(x)]^{i-1} dx$$

$$\begin{aligned} & \left. \begin{aligned} u &= F(x) \\ \frac{du}{dx} &= f(x) \Rightarrow du = f(x) dx \end{aligned} \right\}^* \end{aligned}$$

$$\rightarrow P(Y=i) = \int_0^1 (1-u) u^{i-1} du$$

$$= \int_0^1 u^{i-1} - u^i du$$

$$= \left[\frac{u^i}{i} - \frac{u^{i+1}}{i+1} \right]_0^1$$

$$= \frac{1}{i} - \frac{1}{i+1} = \frac{(i+1) - i}{(i+1)i}$$

$$= \frac{1}{i(i+1)}$$

$$P(Y=y) = \frac{1}{y(y+1)}, \quad Y \in \{1, 2, \dots\}$$

$$(b) E(Y) = \sum_{y=1}^{\infty} \frac{1}{y(y+1)} = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

$$5.6 \quad (a.) \quad \begin{aligned} Z &= X - Y & W &= X \\ Y &= W - Z & X &= W \end{aligned} \quad |J| = \begin{vmatrix} \frac{dx}{dw} & \frac{dx}{dz} \\ \frac{dy}{dw} & \frac{dy}{dz} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 1 & -1 \end{vmatrix} = 1$$

$$f_{Z,W}(z,w) = f_X(w) f_Y(w-z) \cdot 1$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(w) f_Y(w-z) dw$$

$$(b.) \quad \begin{aligned} Z &= XY & W &= X \\ Y &= Z/W & X &= W \end{aligned} \quad |J| = \begin{vmatrix} \frac{dx}{dw} & \frac{dx}{dz} \\ \frac{dy}{dw} & \frac{dy}{dz} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -\frac{z}{w^2} & \frac{1}{w} \end{vmatrix} = \frac{1}{w}$$

$$f_{Z,W}(z,w) = f_X(w) f_Y(z/w) \left| \frac{1}{w} \right|$$

$$f_Z(z) = \int_{-\infty}^{\infty} \left| \frac{1}{w} \right| f_X(w) f_Y(z/w) dw$$

$$(c.) \quad \begin{aligned} Z &= \frac{X}{Y} & W &= X \\ Y &= \frac{W}{Z} & X &= W \end{aligned} \quad |J| = \begin{vmatrix} \frac{dx}{dw} & \frac{dx}{dz} \\ \frac{dy}{dw} & \frac{dy}{dz} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ \frac{1}{z} & -\frac{w}{z^2} \end{vmatrix} = -\frac{w}{z^2}$$

$$f_{Z,W}(z,w) = f_X(w) f_Y(w/z) \left| \frac{w}{z^2} \right|$$

$$f_Z(z) = \int_{-\infty}^{\infty} \left| \frac{w}{z^2} \right| f_X(w) f_Y(w/z) dw$$

$$5.11 \quad E(S^2) = \sigma^2 \geq [E(s)]^2 \text{ by Jensen's Inequality}$$

$\Rightarrow$ Take Square Roots of both sides.

$$\sigma \geq E(s)$$

From Jensen's Inequality and definition of variance $E(x^2) = [E(x)]^2$ only if variance of x is 0. Thus

$$E(x^2) > [E(x)]^2, \text{ unless}$$

$P(X = EX) = 1$. Likewise if $\sigma^2 > 0$, then $\sigma > E(s)$.

9.

$$\begin{aligned} \text{Var}(S^2) &= \text{Var} \left[\frac{(n-1)\sigma^2 S^2}{(n-1)\sigma^2} \right] \\ &= \frac{\sigma^4}{(n-1)^2} \text{Var} \left[\frac{(n-1)S^2}{\sigma^2} \right] \\ &= \frac{\sigma^4}{(n-1)^2} \text{Var}[Y], \text{ where } Y \sim \chi_{n-1}^2 = \text{Gamma} \left(\frac{n-1}{2}, 2 \right) \\ &= \frac{\sigma^4}{(n-1)^2} \frac{n-1}{2} 2^2 \\ &= \frac{2\sigma^4}{(n-1)} \end{aligned}$$

11.

Let $U \sim \chi_p^2$ and $V \sim \chi_q^2$, independent. Their joint pdf is

$$\frac{1}{\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{q}{2}\right)2^{(p+q)/2}}u^{\frac{p}{2}-1}v^{\frac{q}{2}-1}e^{\frac{-(u+v)}{2}}.$$

From Definition 5.3.6, the random variable $X = (U/p)/(V/q)$ has an F distribution, so we make the transformation $x = (u/p)/(v/q)$ and $y = u + v$. (Of course, many choices of y will do, but this one makes calculations easy. The choice is prompted by the exponential term in the pdf.) Solving for u and v yields

$$u = \frac{\frac{p}{q}xy}{1 + \frac{q}{p}x}, \quad v = \frac{y}{1 + \frac{q}{p}x}, \quad \text{and } |J| = \frac{\frac{q}{p}y}{\left(1 + \frac{q}{p}x\right)^2}.$$

We then substitute into $f_{U,V}(u, v)$ to obtain

$$f_{X,Y}(x, y) = \frac{1}{\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{q}{2}\right)2^{(p+q)/2}}\left(\frac{\frac{p}{q}xy}{1 + \frac{q}{p}x}\right)^{\frac{p}{2}-1}\left(\frac{y}{1 + \frac{q}{p}x}\right)^{\frac{q}{2}-1}e^{\frac{-y}{2}}\frac{\frac{q}{p}y}{\left(1 + \frac{q}{p}x\right)^2}.$$

Note that the pdf factors, showing that X and Y are independent, and we can read off the pdfs of each: X has the F distribution and Y is χ_{p+q}^2 . If we integrate out y to recover the proper constant, we get the F pdf

$$f_X(x) = \frac{\Gamma\left(\frac{p+q}{2}\right)}{\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{q}{2}\right)}\left(\frac{q}{p}\right)^{p/2}\frac{x^{p/2-1}}{\left(1 + \frac{q}{p}x\right)^{\frac{p+q}{2}}}.$$

5.16 a. $\sum_{i=1}^3 \left(\frac{X_i - i}{i}\right)^2 \sim \chi_3^2$

b. $\left(\frac{X_i - 1}{i}\right) \bigg/ \sqrt{\sum_{i=2}^3 \left(\frac{X_i - i}{i}\right)^2} \bigg/ 2 \sim t_2$

c. Square the random variable in part b).

13.

- (b) Derive the mean and variance* of X . (* Variance was not required).

Since $F_{p,q} = \frac{\chi_p^2/p}{\chi_q^2/q}$, let $U \sim \chi_p^2$ and $V \sim \chi_q^2$, independent. Then

$$E(F_{p,q}) = E\left(\frac{U/p}{V/q}\right) = E\left(\frac{U}{p}\right) E\left(\frac{q}{V}\right) = \frac{p}{p} q E\left(\frac{1}{V}\right) = q E\left(\frac{1}{V}\right).$$

We can find $E(1/V)$ with

$$E\left(\frac{1}{V}\right) = \int_0^\infty \frac{1}{v} \frac{1}{\Gamma\left(\frac{q}{2}\right) 2^{q/2}} v^{q/2-1} e^{-v/2} dv = \frac{1}{q-2}.$$

Hence $E(F_{p,q}) = \frac{q}{q-2}$. The computation of the variance is similar and involves finding $E(U^2)$ and $E\left(\frac{1}{V^2}\right)$.

- (c) Show that $1/X$ has an $F_{q,p}$ distribution.

Let $X = \frac{U/p}{V/q}$ as in part (a). Then $\frac{1}{X} = \frac{V/q}{U/p}$ which will be $F_{q,p}$ following the derivation in part (a).

- (d) Show that $(p/q)X/[1 + (p/q)X]$ has a beta distribution with parameters $p/2$ and $q/2$.

Let $Y = \frac{(p/q)X}{1+(p/q)X} = \frac{pX}{q+pX}$. Then $X = \frac{qY}{p(1-Y)}$ and $\left|\frac{dx}{dy}\right| = (q/p)(1-y)^{-2}$ and the pdf of Y is

$$\begin{aligned} f_Y(y) &= \frac{\Gamma\left(\frac{q+p}{2}\right)}{\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{q}{2}\right)} \left(\frac{p}{q}\right)^{p/2} \frac{\left(\frac{qy}{p(1-y)}\right)^{(p-2)/2}}{\left(1 + \frac{pqy}{qp(1-y)}\right)^{(p+q)/2}} \frac{q}{p(1-y)^2} \\ &= \frac{\Gamma\left(\frac{q+p}{2}\right)}{\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{q}{2}\right)} y^{(p/2)-1} (1-y)^{(q/2)-1} \\ &\sim \text{Beta}\left(\frac{p}{2}, \frac{q}{2}\right). \end{aligned}$$

CB 5.18 Let X be a random variable with a Student's t distribution with p degrees of freedom.

- (a) Derive the mean and variance of X .

$E(X) = E(Z/\sqrt{V/p}) = E(Z)/E(\sqrt{V/p}) = 0$ since $E(Z) = 0$. $\text{Var}(X) = E(X^2) = p/(p-2)$ since X^2 is $F_{1,p}$.

- (b) Show that X^2 has an F distribution with 1 and p degrees of freedom.

$X^2 = Z^2/(V/p)$. Since $Z^2 \sim \chi_p^2$, this ratio is an $F_{1,p}$.

15.

The given integral is

$$\begin{aligned} &\int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2 x/2} \nu \sqrt{x} \frac{1}{\Gamma(\nu/2) 2^{\nu/2}} (\nu x)^{(\nu/2)-1} e^{-\nu x/2} dx \\ &= \frac{1}{\sqrt{2\pi}} \frac{\nu^{\nu/2}}{\Gamma(\nu/2) 2^{\nu/2}} \int_0^\infty e^{-t^2 x/2} x^{((\nu+1)/2)-1} e^{-\nu x/2} dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{2\pi}} \frac{\nu^{\nu/2}}{\Gamma(\nu/2)2^{\nu/2}} \int_0^\infty x^{((\nu+1)/2)-1} e^{-(\nu+t^2)x/2} dx && \left(\begin{array}{l} \text{integrand is kernel of} \\ \text{gamma}((\nu+1)/2, 2/(\nu+t^2)) \end{array} \right) \\
&= \frac{1}{\sqrt{2\pi}} \frac{\nu^{\nu/2}}{\Gamma(\nu/2)2^{\nu/2}} \Gamma((\nu+1)/2) \left(\frac{2}{\nu+t^2} \right)^{(\nu+1)/2} \\
&= \frac{1}{\sqrt{\nu\pi}} \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)} \frac{1}{(1+t^2/\nu)^{(\nu+1)/2}},
\end{aligned}$$

the pdf of a t_ν distribution.